

2.19

1. a. $(\text{Fun } z \rightarrow \text{Fun } y \rightarrow \text{False and True})$
 b. $(\text{Fun } x \rightarrow \text{Fun } z \rightarrow 5 + z)$
 c. $(\text{Fun } x \rightarrow \text{Fun } z \rightarrow \text{If True Then } x \text{ Else } z)$
 d. $(\text{If True Then } (\text{Fun } z \rightarrow \text{Fun } y \rightarrow y \text{ And } z) \text{ Else } y \text{ And } z)$
 e. $(\text{Fun } x \rightarrow x) (\text{Fun } y \rightarrow 3)$
 f. $(\text{Let } y = x + 1 \text{ In } y + 2)$
 g. $(\text{Let } f = (\text{Fun } f \rightarrow f) \text{ In } f)$

2. a. it's missing the application of (= rule) $\frac{6 \Rightarrow 6 \quad 6 \Rightarrow 6}{6 = 6 \Rightarrow \text{True}}$ and (If True rule).
 b. the step $[2/w]$ should be a no-op since w is bounded in $(\text{Let } w = 3 \text{ In } w + w)$
 $\hookrightarrow$ in the line $\overline{2 \Rightarrow 2} (\text{Let } w = 3 \text{ In } w + w) [2/w]$.
 c. call-by-value, but the argument is not computed $\cdot 3 \Rightarrow 3$.

3.

- a.
$$\text{Fun } z \Rightarrow \text{If } ((\text{Fun } y \Rightarrow \text{Not } y) z) \text{ Then } 1 \text{ Else } 0$$

$$\frac{(\text{Fun } y \Rightarrow \text{Not } y) \Rightarrow \star \quad \text{Fun } z \Rightarrow \text{If } (f z) \text{ Then } 1 \text{ Else } 0 \quad [\text{Fun } y \Rightarrow \text{Not } y / f]}{\text{Let } f = (\text{Fun } y \Rightarrow \text{Not } y) \text{ In Fun } z \Rightarrow \text{If } (f z) \text{ Then } 1 \text{ Else } 0 \Rightarrow \text{Fun } z \Rightarrow \text{If } ((\text{Fun } y \Rightarrow \text{Not } y) z) \text{ Then } 1 \text{ Else } 0}$$

b.

$$\frac{\frac{\frac{\text{False} \Rightarrow \text{False}}{\text{Not False} \Rightarrow \text{True}}}{(\text{Fun } y \Rightarrow \dots) \Rightarrow \star \quad \text{False} \Rightarrow \star \quad \text{Not } y [\text{False} / y]}{((\text{Fun } y \Rightarrow \text{Not } y) \text{ False}) \Rightarrow \text{True} \quad 1 \Rightarrow 1}}{\text{If } ((\text{Fun } y \Rightarrow \text{Not } y) \text{ False}) \text{ Then } 1 \text{ Else } 0 \Rightarrow 1}$$

$$\frac{(\text{by problem 3.a}) \quad \text{False} \Rightarrow \star \quad \text{If } ((\text{Fun } y \Rightarrow \text{Not } y) z) \text{ Then } 1 \text{ Else } 0 [\text{False} / z]}{(\text{Fun } f \Rightarrow \text{Fun } z \Rightarrow \text{If } (f z) \text{ Then } 1 \text{ Else } 0) (\text{Fun } y \Rightarrow \text{Not } y) \text{ False} \Rightarrow 1}$$

c.

$$\frac{\frac{\frac{0 \Rightarrow 0 \quad 0 \Rightarrow 0}{0 = 0 \Rightarrow \text{True}} \quad \text{True} \Rightarrow \text{True}}{\text{If } 0 = 0 \text{ Then True Else Fun } x \Rightarrow x \Rightarrow \text{True}}}{\frac{(\text{Fun } x \Rightarrow \dots) \Rightarrow \star \quad 0 \Rightarrow 0 \quad (\text{If } x = 0 \text{ Then True Else Fun } x \Rightarrow x) [0 / x]}{((\text{Fun } x \Rightarrow \text{If } x = 0 \text{ Then True Else Fun } x \Rightarrow x) 0) \Rightarrow \text{True}}}$$

$$\text{Not } ((\text{Fun } x \Rightarrow \text{If } x = 0 \text{ Then True Else Fun } x \Rightarrow x) 0) \Rightarrow \text{False}$$

4. a. FbEr's BNF: $x ::= \text{Fb's variables}, y ::= \text{Fb's values} \mid \text{Err}, e ::= \text{Fb's expressions}$

- b. (Err Not Rule)

$$\frac{e \Rightarrow \text{Err}}{\text{Not } e \Rightarrow \text{Err}}$$

$$\text{(Err And Rule)} \quad \frac{e_1 \Rightarrow \text{Err} \quad e_2 \Rightarrow v_2}{e_1 \text{ And } e_2 \Rightarrow \text{Err}} \quad \frac{e_1 \Rightarrow v_1 \quad e_2 \Rightarrow \text{Err}}{e_1 \text{ And } e_2 \Rightarrow \text{Err}} \quad \frac{e_1 \Rightarrow \text{Err} \quad e_2 \Rightarrow \text{Err}}{e_1 \text{ And } e_2 \Rightarrow \text{Err}}$$